

NGUYEN MY THONG

Software Engineer - AI Engineer - AI Integration

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SUMMARY

Software Engineer specializing in AI Integration and High-Performance Backend Architecture. Proven ability to architect production-ready Agentic RAG pipelines, scalable Go microservices, and complex ETL workflows. Adept at bridging the gap between advanced AI models (LLMs, Vector DBs, Rerankers) and intuitive user interfaces (SvelteKit, Next.js). Possesses a strong engineering mindset with experience in Clean Architecture, Real-time WebSockets, and System Resilience, delivering impactful solutions for enterprise clients and complex academic projects.

EDUCATION & CERTIFICATIONS

University of Information Technology – UIT

Computer Science student.

Google Developer Student Club UIT

Head Developer of GDSC club at University - UIT

SKILLS

- **Languages:** Python, Go, TypeScript/JavaScript, C++, Java.
- **AI & Data Engineering:** LangGraph, Agentic RAG, Hybrid Search (Dense/Sparse), Cross-Encoder Reranking, SentenceTransformers, HuggingFace, DeepEval, N8N.
- **Backend & Infrastructure:** Go (Gin), FastAPI, WebSockets, Server-Sent Events (SSE), Clean Architecture, Docker.
- **Databases & BaaS:** Qdrant (Vector DB), ChromaDB, PostgreSQL, MongoDB, Directus (Headless CMS), Supabase.
- **Frontend:** Svelte/SvelteKit, Next.js/React.

WORK EXPERIENCE

Visual Hive (UK-based Startup) - Software Engineer (Contract)

August 2025 - Feb 2026

- Spearheaded the Data Ingestion & ETL Pipeline: Engineered a high-performance ETL platform using SvelteKit to replace legacy Noodl applications. Implemented robust CSV/XLSX parsing, fuzzy-matching algorithms for dynamic schema mapping, and 10+ automated data cleaning stages handling thousands of records.
- Architected "Hosted Buyers" Matchmaking Engine: Designed and implemented a complex bi-directional matchmaking algorithm for B2B events. Combined AI-driven semantic similarity scores with mutual user-interest logic to rank matches, integrated with an automated appointment scheduling system.
- Headless CMS & Vector DB Integration: Handled seamless bi-directional synchronization between Directus (Relational DB) and Qdrant (Vector DB). Optimized bulk mutation logic and ensured 100% data consistency for the RAG-based AI conference assistant.
- AI Analytics SaaS Dashboard: Developed a comprehensive analytics dashboard featuring real-time data visualization (LayerChart/D3.js) to track AI KPIs, engagement funnels, evaluation score trends, and conversation pain points.

UIT InSecLab - AI Researcher - Backend Engineer - System Designer

June 2025 - September 2025

- Autonomous Multi-Agent Penetration Testing System: Architected a state-of-the-art LLM-based autonomous agent framework (Collector, Exploiter, Remediator) capable of executing end-to-end security audits for Web2 infrastructure and Web3 Smart Contracts.
- Self-Healing Exploit Generation: Engineered an intelligent refinement loop integrating Foundry. The agent autonomously generates Solidity exploit payloads, executes forge test in isolated environments, analyzes AST/compilation errors, and iteratively fixes its own code until the exploit succeeds.
- Remote Execution & AST Parsing: Developed a robust SSH-based Remote Shell Manager (Paramiko) to orchestrate Kali Linux tools (Slither, Metasploit, Nmap). Implemented complex Regex and AST parsers to extract, sanitize, and structure raw terminal outputs for LLM reasoning.
- LLM Agnostic & State Management: Integrated multiple LLM providers (AWS Bedrock, OpenAI, Gemini, local Ollama via ngrok) with a MySQL (SQLAlchemy) backend to maintain persistent task planning, execution states, and conversational memory across agent lifecycles.

REMARKABLE PROJECTS

SkillForge

March 2025 - May 2025

A full-stack microservices-based platform connecting students with business projects via AI matching.

- **Full-Stack Architecture:** Developed a highly reactive frontend using SvelteKit and architected a robust backend using Go (Gin) and MongoDB, strictly following Clean Architecture principles (Handler-Service-Repository pattern).
- **AI Matching Microservice:** Built a decoupled Python/FastAPI microservice utilizing SentenceTransformers to calculate semantic cosine similarity scores between student profiles and enterprise project requirements.
- **Real-time Collaboration:** Implemented a scalable WebSocket infrastructure using Gorilla WebSocket, seamlessly integrated with the Svelte UI to handle real-time live chat, dynamic Kanban board updates, and instant system notifications.
- **Gamification System:** Developed complex backend logic for dynamic badge awarding and experience point (XP) scaling to increase student engagement.

ViLeXa - Vietnamese Legal Expert Assistant

Dec 2025 - Jan 2026

An advanced Agentic RAG system built to navigate and query complex Vietnamese legal documents.

- **Agentic RAG Architecture:** Engineered an 8-node evaluation pipeline using FastAPI and LangGraph with dynamic query routing, contextual profile updating, and progressive filter relaxation.
- **Advanced Vector Search:** Implemented Hybrid Search (Dense + Sparse) with BGE-M3 multilingual embeddings in Qdrant, coupled with a Cross-Encoder Reranker to optimize retrieval precision for Vietnamese legal terminology.
- **Measurable Impact & Resilience:** Achieved an outstanding 82.7% Hit Rate@1 on the Zalo AI Legal Retrieval benchmark. Designed a production-ready Server-Sent Events (SSE) streaming API featuring concurrency control, Rate Limiting, and Circuit Breakers, verified by DeepEval and Locust load testing to maintain <12s p95 response times under heavy concurrent loads.

JARVIS - Vietnamese AI Voice Assistant

March 2026 - April 2026

A proactive, hardware-integrated AI voice assistant designed for seamless autonomous task management.

- Engineered a personalized AI voice assistant bridging hardware (ESP32) and a custom Python WebSocket server.
 - Integrated Gemini Live native audio pipeline and implemented 20+ function-calling tools for autonomous tasks (reminders, finance, health tracking).
 - Designed a 5-layer persistent memory system using ChromaDB with semantic search for proactive and context-aware user notifications.
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